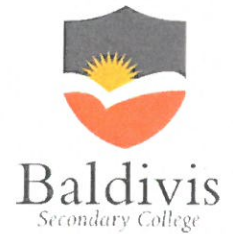


**Year 10
Pathway 2
Mini Test 4
Probability**



Name: SOLUTIONS

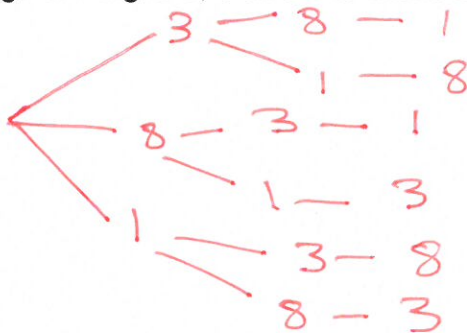
Score: /50

Total Time: 50 minutes

**Full working out must be shown to get full marks.
Attempt all questions**

**Resources allowed:
1 A4 page, (1 side) of hand written notes
Calculator**

1. Arrange the digits 3, 8 and 1 to make all as many different whole numbers as possible. (2,1, 1,1- 5 marks)

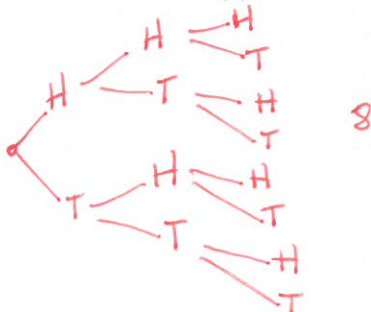


6

Calculate the following.

- a) P(starts with a 8) $\frac{2}{6} = \frac{1}{3}$
 b) P(has 3 after the 1) $\frac{2}{6} = \frac{1}{3}$
 c) P(ends in a 1) $\frac{2}{6} = \frac{1}{3}$

2. Three coins are flipped. List the sample space. (2, 6- 8 marks)



HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

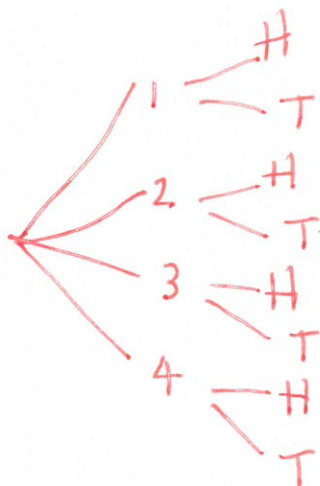
Calculate the probability of

- a) 3 tails. $\frac{1}{8}$
 b) 1 head. $\frac{3}{8}$
 c) no more than 2 heads. $\frac{3}{8}$
 d) all the same $\frac{2}{8} = \frac{1}{4}$
 e) no tails $\frac{1}{8}$
 f) all different $\frac{0}{8}$

3. A four sided die numbered 1 to 4 is rolled and a coin is flipped.

a) Draw a tree diagram of this situation.

(3 Marks)



b) List the sample space.

(1 mark)

$$S = \{1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T\}$$

c) Calculate the probability of

(4 Marks)

(i) a four with a head

$$\frac{1}{8}$$

(iii) a multiple of two. $\frac{2}{4} = \frac{1}{2}$

(ii) a tail

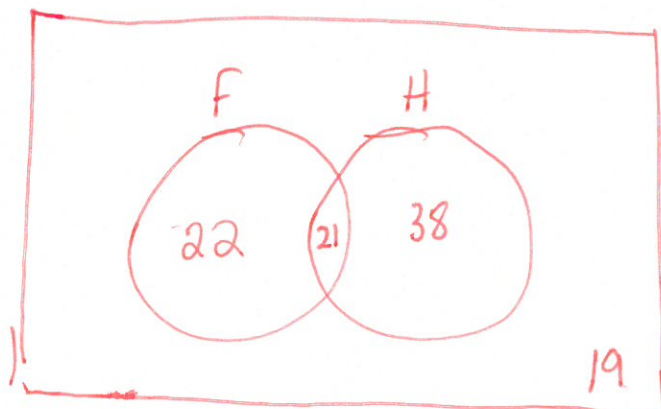
$$\frac{4}{8} = \frac{1}{2}$$

(iv) a tail given an odd number $\frac{2}{4} = \frac{1}{2}$

4. One hundred students are asked about their favourite type of car. Forty three liked Ford and twenty one liked Ford (F) and Holden (H) . Nineteen liked neither.

a) Draw a Venn diagram of this.

(4 Marks)



b) Use the Venn Diagram to calculate the probability of:
 {NB: liking Holden=(H) & liking Ford= (F)}

(4 Marks)

(i) liking Holden $\frac{59}{100}$

(iii) $P(H') = \frac{62}{100} = \frac{31}{50}$

(ii) Liking ford only $\frac{22}{100} = \frac{11}{50}$

(iv) $P(H \cup F) = \frac{19}{100}$
 Not H or F

5. 77 people were asked about their home and some of its features. The following table shows some of the information.

a) Complete the table.

(6 Marks)

	3 bedrooms	4 bedrooms	Total
1 bathroom	17	26	43
2 bathrooms	12	22	34
Total	29	48	77

b) Calculate the following.

(6 Marks)

(i) How many people own a 4 bedroom house?

48

(ii) How many people do not own a house with 3 bathrooms

29

(iii) $P(\text{own 4 bedroom house})$

$\frac{48}{77}$

(iv) $P(\text{own a 3 bedroom given it has 2 bathrooms})$

$\frac{12}{34}$

$\frac{39}{34}$

(v) $P(\text{it has one bathroom and four bedrooms})$

$\frac{26}{77}$

(vi) $P(\text{it did not have four bedrooms and it did not have two bathrooms})$

$\frac{17}{77}$

6. Are these sets mutually exclusive? Explain why.

(2 Marks)

- a) A – 0, 1, 3, 5, 7, 8, 9
B – 2, 4, 6, 9, 11, 12

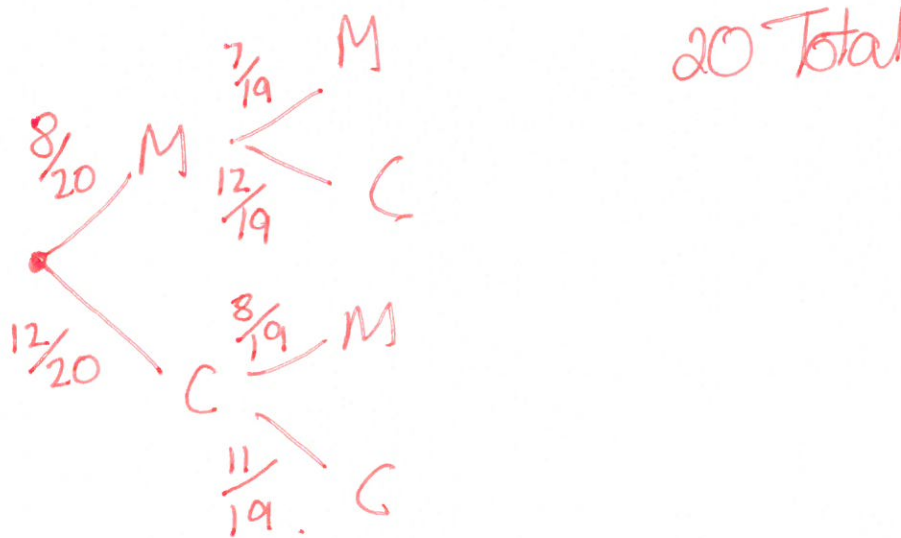
Not Mutually exclusive as
9 appears in both sets

- b) C – a, d, f, j, l, p, q, c
D – w, e, t, u, m, b, x, z

Yes this is mutually exclusive as
there are not letter that appears in
both sets

7 A bag contains eight Minties and twelve Caramels. Two lollies are selected, one after the other, with no replacement. (3, 1. 1-5 Marks)

(a) Draw a tree diagram labelled with probabilities to illustrate this information.



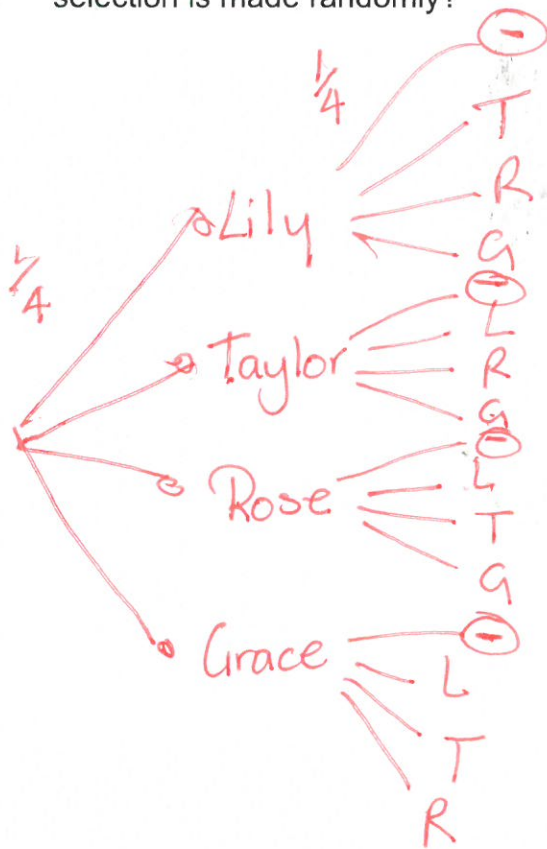
(b) Calculate the probability that the lollies are both Minties.

$$\frac{8}{20} \times \frac{7}{19} = \frac{56}{380} = \frac{28}{190} = \frac{14}{95}$$

(c) Calculate the probability that the lollies are different.

$$\frac{8}{20} \times \frac{12}{19} + \frac{12}{20} \times \frac{8}{19} = \frac{96}{380} + \frac{96}{380} = \frac{192}{380} = \frac{96}{190} = \frac{48}{95}$$

- 8 Mark and Amy are trying to decide on a name for their new baby girl. They like the names Lily, Taylor, Rose and Grace but they don't want to choose more than two names – one name would also be okay. What is the probability that the baby will be named Taylor Grace, if the selection is made randomly? (2 Marks)



16

$\frac{1}{16}$

End of Test

